

Requirement Modeling

CE418: Systems Analysis and Design

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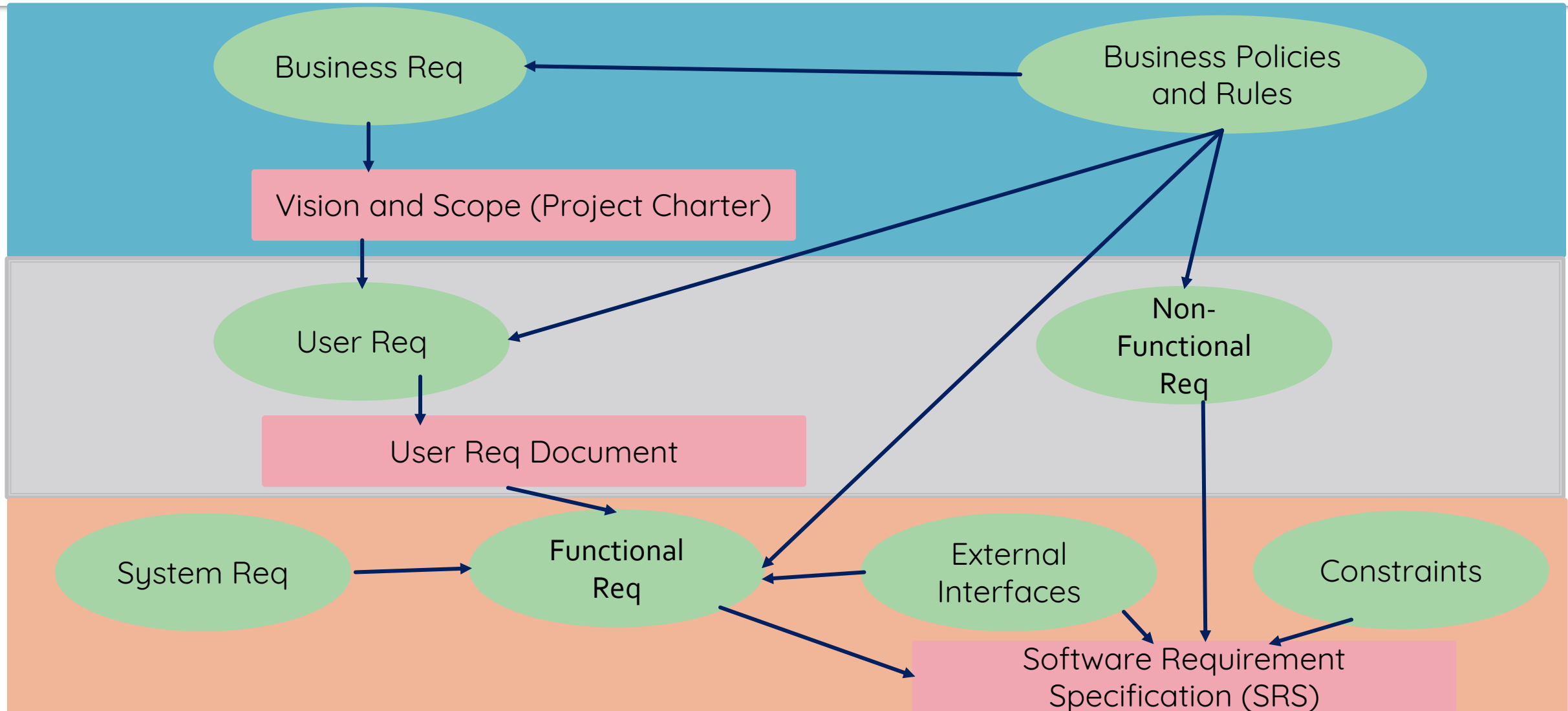
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Introduction

Overview



Requirements Discovery

- **Requirements Discovery**: The process and techniques used by system analysts to identify or extract system problems and solution requirements from the user community.
- **System Requirement**: Something that the information system must do or a property that it must have. Also called a business requirement.

Functional vs Nonfunctional

- Functional Requirement: Something the information system must do.
- Nonfunctional Requirement: A property or quality the system must have:
 - Performance
 - Security
 - Costs

Samples

نیازمندی‌های کارکردی

نیازمندی‌های کارکردی عملکرد و عملیات خاصی را که نرم‌افزار باید انجام دهد، توصیف می‌کنند. این نیازمندی‌ها بر روی اقدامات و فعالیت‌هایی که سیستم باید انجام دهد، تمرکز دارند، مانند محاسبات، پردازش داده‌ها و تعامل با کاربران یا سایر سیستم‌ها. نمونه‌ای از نیازمندی‌های کارکردی عبارتند از:

▲ احراز هویت کاربران

کاربران باید بتوانند با استفاده از نام کاربری و رمز عبور وارد سیستم شوند

▲ ورود اطلاعات

سیستم باید به کاربران اجازه ورود اطلاعات به فیلدهای تعریف شده را بدهد

▲ تولید گزارش

سیستم باید گزارش‌های ماهانه فروش را به صورت پی‌دی‌اف تولید کند

نیازمندی‌های غیرکارکردی

نیازمندی‌های غیرکارکردی، گاهی به عنوان "ویژگی‌های کیفی" یا "محدودیت‌ها" نیز شناخته می‌شوند، ویژگی‌ها یا ویژگی‌هایی را تعیین می‌کنند که سیستم باید داشته باشد یا نشان دهد. این نیازمندی‌ها بر روی نحوه عملکرد سیستم تمرکز دارند تا اینکه سیستم چه کاری انجام می‌دهد. نمونه‌هایی از نیازمندی‌های غیرکارکردی عبارتند از:

▲ عملکرد

سیستم باید به درخواست‌های کاربر در دو ثانیه زمان بیشینه در شرایط بار مشترک پاسخ دهد

▲ امنیت

رمزهای عبور کاربر باید با استفاده از الگوریتم‌های رمزنگاری استاندارد صنعتی رمزگذاری شوند

▲ استفاده‌پذیری

رابط کاربری سیستم باید ساده و آسان برای ملاحظه و مرور با برچسب‌ها و پیام‌های خطا واضح باشد

تفاوت نیازمندی‌های کارکردی و غیرکارکردی

- تمرکز: الزامات عملکردی بر آنچه سیستم انجام می‌دهد تمرکز می‌کنند، در حالی که الزامات غیر عملکردی بر نحوه عملکرد سیستم تمرکز می‌کنند
- ماهیت: الزامات عملکردی اعمال یا رفتارهای خاصی هستند که سیستم باید انجام دهد، در حالی که الزامات غیرعملکردی کیفیت‌ها یا ویژگی‌هایی هستند که سیستم باید داشته باشد
- ملموس بودن: الزامات عملکردی اغلب صریح و به راحتی قابل مشاهده هستند، در حالی که الزامات غیرعملکردی اغلب ضمنی هستند و ممکن است نیاز به تفسیر یا توضیح داشته باشند
- قابلیت اندازه‌گیری: الزامات عملکردی را می‌توان به راحتی بر حسب اینکه آیا آنها به درستی اجرا می‌شوند یا نه اندازه‌گیری کرد، در حالی که الزامات غیر عملکردی ممکن است به ارزیابی ذهنی یا معیارهای خاصی برای ارزیابی نیاز داشته باشند

Results of Incorrect Requirements

- The system may cost more than projected.
- The system may be delivered later than promised.
- The system may not meet the users' expectations and they may not use it.
- Once in production, costs of maintaining and enhancing the system may be excessively high.
- The system may be unreliable and prone to errors and downtime.
- Reputation of IT staff is tarnished as failure will be perceived as a mistake by the team.

Relative Cost to Fix An Error

Phase in Which Error Discovered	Cost Ratio
Requirements	1
Design	3–6
Coding	10
Development Testing	15–40
Acceptance Testing	30–70
Operation	40–1000

Criteria For System Requirements

- **Consistent** – not conflicting or ambiguous.
- **Complete** – describe all possible system inputs and responses.
- **Feasible** – can be satisfied based on the available resources and constraints.
- **Required** – truly needed and fulfill the purpose of the system.
- **Accurate** – stated correctly.
- **Traceable** – directly map to functions and features of the system.
- **Verifiable** – defined so it can be demonstrated during testing.

Requirements Discovery

Requirements Discovery

Given an understand of problems, the systems analyst can start to define requirements.

Fact-finding – the formal process of using research, meetings, interviews, questionnaires, sampling, and other techniques to collect information about system problems, requirements, and preferences. It is also called *information gathering* or *data collection*.

Software Requirements Specification (SRS)

- A formal document that communicates the requirements of a proposed system to key stakeholders and serves as a contract for the systems project.
- Synonyms:
 - Requirements Definition Report
 - Requirements Statement
 - Requirements Definition Document
 - Functional Specifications

Sample SRS

REQUIREMENTS DEFINITION REPORT

1. Introduction
 - 1.1. Purpose
 - 1.2. Background
 - 1.3. Scope
 - 1.4. Definitions, Acronyms, and Abbreviations
 - 1.5. References
 2. General Project Description
 - 2.1. Functional Requirements
 3. Requirements and Constraints
 - 3.1. Functional Requirements
 - 3.2. Nonfunctional Requirements
 4. Conclusion
 - 4.1. Outstanding Issues
- Appendix (optional)

Requirements Management

- **Requirements management** - the process of managing change to the requirements.
 - Over the lifetime of the project it is very common for new requirements to emerge and existing requirements to change.
 - Studies have shown that over the life of a project as much as 50 percent or more of the requirements will change before the system is put into production.

Requirements Discovery Methods

Seven Fact-Finding Methods

- Sampling of existing documentation, forms, and databases
- Research and site visits
- Observation of the work environment
- Questionnaires
- Interviews
- Prototyping
- Joint requirements planning (JRP).

Sampling of existing documentation, forms, and databases

- **Sampling** –process of collecting a representative sample of documents, forms, and records.
 - Organization chart
 - Memos and other documents that describe the problem
 - Standard operating procedures for current system
 - Completed forms
 - Manual and computerized screens and reports
 - Samples of databases
 - Flowcharts and other system documentation
 - And more

Sample Completed rather than Blank

- Can determine type of data going into each blank

Can determine size of data going into each blank

Can determine
which blanks
are not used
or not always
used

Can see data
relationships

IT SERVICES

Service Request



Computer Name Reverb	Assigned To CIS	Report Date 2/18/01	Resolution Date 3/1/01
Reported By Greg Kaufman		Ext 0982	

Problem Description
Monitor won't come on

Work Date	Tech	Work Comments
2/18/01	Connie Bailey	Verified problem in video card. Installed loaner card
3/1/01	Connie Bailey	Installed new video card

Observation

Observation – a fact-finding technique wherein the systems analyst either participates in or watches a person perform activities to learn about the system.

Advantages?

Disadvantages?

Work sampling - a fact-finding technique that involves a large number of observations taken at random intervals.

Questionnaires

- **Questionnaire** – a special-purpose document that allows the analyst to collect information and opinions from respondents.

Free-format questionnaire – a questionnaire designed to offer the respondent greater latitude in the answer. A question is asked, and the respondent records the answer in the space provided after the question.

Fixed-format questionnaire – a questionnaire containing questions that require selecting an answer from predefined available responses.

Interview

Interview - a fact-finding technique whereby the systems analysts collect information from individuals through face-to-face interaction.

- Find facts
- Verify facts
- Clarify facts
- Generate enthusiasm
- Get the end-user involved
- Identify requirements
- Solicit ideas and opinions

The personal interview is generally recognized as the most important and most often used fact-finding technique.

Discovery Prototyping

- **Discovery prototyping** – the act of building a small-scale, representative or working model of the users' requirements in order to discover or verify those requirements.

Joint Requirements Planning (JRP)

- **Joint requirements planning (JRP)** – a process whereby highly structured group meetings are conducted for the purpose of analyzing problems and defining requirements.
 - JRP is a subset of a more comprehensive joint application development or JAD technique that encompasses the entire systems development process.

Any Question?